WORKED SOLUTION

STEP III 2012, Question 8

Pure • Fibonacci Telescoping Series

STEP III 2012, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q8

The key idea

The expression in (ii) changes sign every time n goes up by one, and part (i) is a nudge to look for exactly that kind of pattern. Once we know its value for odd n, the arctan identity in (iii) is just the $\tan(A-B)$ formula in disguise. Rearranged, that identity writes each term of the series as a *difference* of two arctans, so the sum telescopes.

Solution

The first few terms are

$$F_0, F_1, \ldots, F_8 = 0,\ 1,\ 1,\ 2,\ 3,\ 5,\ 8,\ 13,\ 21.$$

Part (i)

Using these values,

$$F_0F_3 - F_1F_2 = 0\cdot 2 - 1\cdot 1 = -1, \qquad F_2F_5 - F_3F_4 = 1\cdot 5 - 2\cdot 3 = -1,$$

so the two expressions are equal, as required.

Part (ii)

Let $G_n = F_nF_{n+3} - F_{n+1}F_{n+2}$. Part (i) says $G_0 = G_2$, which suggests comparing G_{n+1} with G_n. Using the recurrence in the form $F_{n+4} = F_{n+3} + F_{n+2}$,

$$\begin{aligned} G_{n+1} &= F_{n+1}F_{n+4} - F_{n+2}F_{n+3} \\ &= F_{n+1}F_{n+3} + F_{n+1}F_{n+2} - F_{n+2}F_{n+3} \\ &= F_{n+1}F_{n+2} - F_{n+3}\left(F_{n+2} - F_{n+1}\right) \\ &= F_{n+1}F_{n+2} - F_{n+3}F_n \\ &= -G_n, \end{aligned}$$

where in the fourth line we used the recurrence again, $F_{n+2} - F_{n+1} = F_n$.

So G_n changes sign at each step. Since $G_0 = -1$ from part (i), it follows (by induction on n) that $G_n = (-1)^{n+1}$. The two cases are

$$F_n F_{n+3} - F_{n+1}F_{n+2} = \begin{cases} -1 & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

For example, $n = 1$ gives $1 \cdot 3 - 1 \cdot 2 = 1$, and $n = 4$ gives $3 \cdot 13 - 5 \cdot 8 = -1$.

Part (iii)

Fix $r \geq 1$ and let

$$\alpha = \arctan \frac{1}{F_{2r}}, \qquad \beta = \arctan \frac{1}{F_{2r+1}}.$$

All the F_k with $k \geq 1$ are positive integers, so these are well defined. By the compound-angle formula,

$$\tan(\alpha - \beta) = \frac{\dfrac{1}{F_{2r}} - \dfrac{1}{F_{2r+1}}}{1 + \dfrac{1}{F_{2r}F_{2r+1}}} = \frac{F_{2r+1} - F_{2r}}{F_{2r}F_{2r+1} + 1} = \frac{F_{2r-1}}{F_{2r}F_{2r+1} + 1},$$

using $F_{2r+1} - F_{2r} = F_{2r-1}$. Now apply part (ii) with the *odd* value $n = 2r - 1$:

$$F_{2r-1}F_{2r+2} - F_{2r}F_{2r+1} = 1, \qquad \text{so} \qquad F_{2r}F_{2r+1} + 1 = F_{2r-1}F_{2r+2}.$$

Since $F_{2r-1} \geq 1$, it is non-zero and cancels:

$$\tan(\alpha - \beta) = \frac{F_{2r-1}}{F_{2r-1}F_{2r+2}} = \frac{1}{F_{2r+2}}.$$

Ask yourself Equal tangents don't always mean equal angles. Why may we conclude $\alpha - \beta = \arctan(1/F_{2r+2})$?

Because tan is one-to-one on $(-\frac{\pi}{2}, \frac{\pi}{2})$, it is enough to check that both angles lie in that interval. Here $0 < \beta \leq \alpha \leq \frac{\pi}{4}$, because $1 \leq F_{2r} \leq F_{2r+1}$, so $0 \leq \alpha - \beta < \frac{\pi}{4}$. Also $\arctan(1/F_{2r+2})$ lies in $(0, \frac{\pi}{2})$ by definition. Hence

$$\arctan \frac{1}{F_{2r}} = \arctan \frac{1}{F_{2r+1}} + \arctan \frac{1}{F_{2r+2}} \qquad (r = 1, 2, 3, \ldots),$$

which is the required identity.

The sum. Rearranged, the identity reads

$$\arctan \frac{1}{F_{2r+1}} = \arctan \frac{1}{F_{2r}} - \arctan \frac{1}{F_{2r+2}},$$

so the partial sums telescope:

$$\begin{aligned}\sum_{r=1}^{N} \arctan\frac{1}{F_{2r+1}} &= \left(\arctan\frac{1}{F_2} - \arctan\frac{1}{F_4}\right) + \left(\arctan\frac{1}{F_4} - \arctan\frac{1}{F_6}\right) + \cdots \\ &\quad + \left(\arctan\frac{1}{F_{2N}} - \arctan\frac{1}{F_{2N+2}}\right) \\ &= \arctan\frac{1}{F_2} - \arctan\frac{1}{F_{2N+2}} = \frac{\pi}{4} - \arctan\frac{1}{F_{2N+2}},\end{aligned}$$

since $F_2 = 1$. For $n \geq 2$ we have $F_{n+1} = F_n + F_{n-1} \geq F_n + 1$, so $F_n \to \infty$. Then $1/F_{2N+2} \to 0$, and since arctan is continuous with $\arctan 0 = 0$, letting $N \to \infty$ gives

$$\sum_{r=1}^{\infty} \arctan\frac{1}{F_{2r+1}} = \frac{\pi}{4}.$$

Checks

- $r = 1$: the identity says $\arctan 1 = \arctan\frac{1}{2} + \arctan\frac{1}{3}$. This is the classic $\frac{\pi}{4}$ identity, since tan of the right-hand side is $\left(\frac{1}{2} + \frac{1}{3}\right)/\left(1 - \frac{1}{6}\right) = 1$.
- The first three terms $\arctan\frac{1}{2} + \arctan\frac{1}{5} + \arctan\frac{1}{13} \approx 0.4636 + 0.1974 + 0.0768 = 0.7378$, which matches $\frac{\pi}{4} - \arctan\frac{1}{21} \approx 0.7378$.
- Numerically, 89 terms of the series give $0.785398163\ldots$, agreeing with $\frac{\pi}{4}$ to machine precision. The formula of part (ii) also holds exactly for every $n < 150$.

Common slips

- Using the even case of (ii) in part (iii). The indices there are $2r - 1$, $2r$, $2r + 1$, $2r + 2$, so $n = 2r - 1$ is odd and the value is $+1$.
- Going from equal tangents to equal angles without checking that both angles lie in $(-\frac{\pi}{2}, \frac{\pi}{2})$.
- Forgetting the first term of the telescoped sum: it is $\arctan(1/F_2) = \frac{\pi}{4}$, not $\arctan(1/F_1)$.

Beyond STEP

Part (ii) is a cousin of Cassini's identity $F_{n-1}F_{n+1} - F_n^2 = (-1)^n$. Both are best understood through the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, whose nth power has the Fibonacci numbers as entries, so that taking determinants gives Cassini's identity at once. Telescoping series, and the care needed to pass from partial sums to a limit, are a staple of a first analysis course. Abbott's *Understanding Analysis* is a good place to read about them, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with recurrences.

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